MIL-T-27493A (ASG)
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MILITARY SPECIFICATION

TRANSFORMER, VARIABLE, SINGLE PHASE, 400 CYCLES
GENERAL SPECIFICATION FOR

This specification has been approved by the Department of the Air Force and by the Bureau of Naval Weapons.

1. SCOPE

1.1 This specification covers the general requirements for tapped or continuously variable transformers (see 6.3.1) for aircraft light dimming controls energized from 400-cycle power sources.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MS33586	Metals, Definition of Dissimilar
MS33742	Shafts, Control Knob

FSC 5950

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H28 Screw-Thread Standards for Federal Services

(Application for copies of the above publication should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Preproduction.- This specification provides for preproduction testing.

3.2 Data.- Unless otherwise specified in the contract or order, no data are required by this specification or any of the documents referenced in section 2 (see 6.2).

3.3 Selection of specifications and standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials.-

3.4.1 Fungus-proof materials.- Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity.

3.4.2 Metals.- Metals shall be corrosion resistant or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions as may be encountered in storage or normal service.

3.4.3 Dissimilar metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other.

3.5 Design and construction.- The design of the transformer shall be in accordance with the applicable military standard (MS) or drawing (see 6.2). The transformer shall be so designed and constructed that no parts will work loose in service. The transformer shall withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.5.1 Reliability.- The transformer shall be so designed and constructed as to ensure the highest degree of operational reliability under the conditions specified herein.

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3.5.2 Voltage control.- The transformer shall incorporate a voltage control that will provide the range of output voltages from the "off" position to its maximum voltage as specified on the applicable MS or drawing. Where tap switching is employed, a minimum of 10 taps shall be provided.

3.5.3 Control rotation.- The full angular rotation of the control shaft shall be 300 degrees ±10 degrees. Where 10 switch taps are used, the angle between adjacent tap centers shall be 30 degrees ±1 degree. The "off" position shall be at the low voltage side of the control at the extreme counterclockwise position of the shaft.

3.5.3.1 Stops.- Stops shall be provided at each end of the shaft rotation.

3.5.4 Terminals.- Stud terminals shall be provided as shown on the applicable MS or drawing. Terminals shall be marked HV (for input voltage), LV (for output voltage), and G (for common ground).

3.5.5 Mounting.- The transformer shall be designed for mounting on a panel by the single-hole method. A nonturn indexing pin shall be provided as shown on the applicable MS or drawing.

3.5.6 Insulation.- The circuit of the transformer shall be electrically insulated from its housing.

3.5.7 Control knob shaft.- The control knob shaft shall conform to MS35742 for a nominal size of 0.250 inch.

3.6 Performance.-

3.6.1 Operation.- When tested as specified in 4.7.1, the transformers shall meet the requirements specified herein and on the applicable MS or drawing without electrical or mechanical damage.

3.6.2 Voltage regulation.- When tested as specified in 4.7.2, the change in the output voltage from a no-load condition to the rated maximum resistive load shall not exceed 10 percent for all output settings of the transformer.

3.6.3 Stop strength.- When tested as specified in 4.7.3, the stops shall withstand 25 pound-inches of torque applied to the shaft without any indication of damage.

3.6.4 Shaft torque.- When tested as specified in 4.7.4, the torque to prevent self-rotation of the shaft shall not exceed 5 pound-inches.

3.6.5 Terminal torque.- When tested as specified in 4.7.5, the terminals shall withstand 5 pound-inches of torque without permanent deflection or failure.

3.6.6 Dielectric strength.- When tested as specified in 4.7.6, the wiring shall withstand a potential of 1,000 volts (V) root mean square (rms), 60 cycles per second (cps) for 1 minute without any evidence of insulation breakdown.

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3.6.7 Overload rating.- When tested as specified in 4.7.7, the transformer shall withstand without permanent distortion or damage 250 percent overload for a cyclic period of 1 minute "on" and 19 minutes "off" or 150 percent overload for a cyclic period of 5 minutes "on" and 15 minutes "off" for 100 hours.

3.6.8 Life.- When tested as specified in 4.7.8, the transformer shall be capable of 500 hours of continuous operation at 52° C ambient without failure at any voltage and current output within its range.

3.6.9 Cycling.- When tested as specified in 4.7.9, the transformer shall be capable of 10,000 operational cycles at rated load (see 6.3.3). There shall be no evidence of damage to the shaft or the transformer.

3.6.10 Temperature rise.- When tested as specified in 4.7.10, with the transformer delivering its maximum rated output, the maximum temperature rise shall not exceed 100° C above an ambient temperature of 25° C.

3.6.11 Volt-ampere output.- When tested as specified in 4.7.11, the transformer shall be capable of delivering the percentage of the maximum current into a fixed resistive load within the range of output voltages corresponding to the curve shown on figure 1. At any point above 30 percent of the maximum output voltage, the departure from this curve shall not exceed 5 percent. Output current shall not be interrupted during any change of the voltage control.

3.6.11.1 The percentage of maximum output voltage for each tap shall be within ±5 percent of the following values:

Tap No.	Percentage maximum voltage
1	31.0
2	36.7
3	41.4
4	46.0
5	52.0
6	58.2
7	66.5
8	75.0
9	87.5
10	100.0

3.6.12 Temperature.- When tested as specified in 4.7.12.1 and 4.7.12.2, there shall be no evidence of mechanical or electrical damage to the transformer and it shall be operative at the conclusion of the test.

3.6.13 Humidity.- When tested as specified in 4.7.12.3, the transformer, at the completion of the test, shall show no signs of excessive corrosion and then shall withstand the dielectric strength test of 4.7.6 using a potential of 500V in lieu of 1,000V.

3.6.14 Altitude.- When tested as specified in 4.7.12.4, the transformer shall withstand any pressure between sea level and 80,000 ft. and the temperature of the windings, as determined by the resistance method, shall not exceed 125° C.

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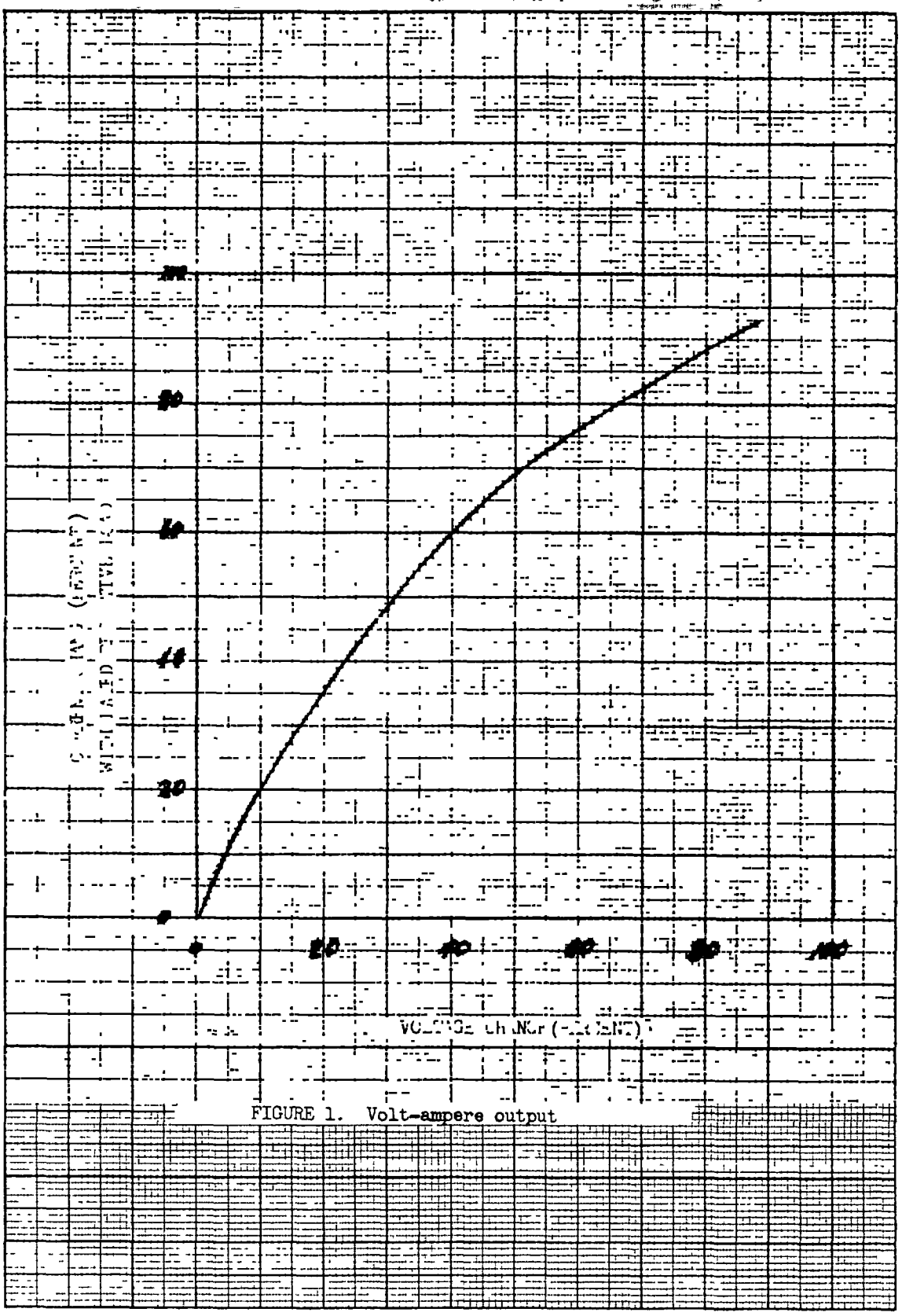


FIGURE 1. Volt-ampere output

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3.6.15 Vibration.- When tested as specified in 4.7.12.5, there shall be no evidence of mechanical or electrical damage and the transformer shall be operative at the conclusion of the test.

3.6.16 Acceleration.- When tested as specified in 4.7.12.6, there shall be no evidence of mechanical or electrical damage and the transformer shall be operative at the conclusion of the test.

3.6.17 Fungus.- When tested as specified in 4.7.12.7, the transformer shall show no appreciable growth of fungi and shall be operative at the conclusion of the test.

3.6.18 Sand and dust.- When tested as specified in 4.7.12.8, there shall be no evidence of mechanical or electrical damage and the transformer shall be operative at the conclusion of the test.

3.6.19 Temperature shock.- When tested as specified in 4.7.12.9, there shall be no evidence of mechanical or electrical damage and the transformer shall be operative at the conclusion of the test.

3.6.20 Explosion-proof.- When tested as specified in 4.7.12.10, there shall be no evidence of mechanical or electrical damage and the transformer shall be operative at the conclusion of the test.

3.6.21 Salt spray.- When tested as specified in 4.7.12.11, there shall be no evidence of mechanical or electrical damage.

3.6.22 Radio interference.- When tested as specified in 4.7.13, the transformer shall meet the interference tests of MIL-I-6181.

3.7 Interchangeability.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-70327 shall govern changes in the manufacturer's part numbers.

3.8 Dimensions.- The dimensions of the transformer shall be as specified on the applicable MS or drawing.

3.9 Weight.- The weight of the transformer shall not exceed the value shown on the applicable MS or drawing. Every practical means shall be taken to hold the weight to the minimum.

3.10 Identification of product.- Equipment, assemblies, and parts shall be marked for [illegible] in accordance with MIL-STD-130.

3.11 Screw threads.- Unless otherwise specified, the threads of all machine screws, 0.06 inch or larger in diameter, shall conform to Handbook H28.

3.12 Workmanship.- The transformer, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

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3.12.1 Screw assemblies.- Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage to the screw or bolt or threads.

3.12.2 Riveting.- Riveting operations shall be performed to assure that the rivets are tight and satisfactorily headed.

3.12.3 Cleaning.- The transformer shall be thoroughly cleaned of loose, spattered, or excess solder; metal chips; and other foreign material after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

3.12.4 Dimensional tolerances.- Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the transformer, they shall be held or limited accordingly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities.- Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government.

4.2 Classification of inspection.- The examining and testing of the transformer shall be classified as:

(a) Preproduction inspections (4.4)
(b) Quality conformance inspections (4.5)

4.3 Test conditions.- Unless otherwise specified, the transformers shall be tested at an ambient temperature of approximately 25° C, at normal atmospheric pressure (approximately 29.92 inches Hg.), at rated voltage and frequency ±3 percent tolerance, and with a rated continuous current lamp load.

4.4 Preproduction inspections.-

4.4.1 Sampling.- The preproduction test samples shall consist of three transformers representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

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4.4.2 Inspections.- Preproduction inspections shall consist of all the examinations and tests specified in 4.6 and 4.7, and shall be conducted as follows:

Sample No. 1	Sample No. 2	Sample No. 3
Examination of product	Examination of product	Examination of product
Dielectric strength	Dielectric strength	Dielectric strength
Shaft torque	Shaft torque	Shaft torque
Operation	Operation	Operation
Voltage regulation	Voltage regulation	Voltage regulation
High temperature	Stop strength	Shock
Low temperature	Cycling operation	Acceleration
Thermal shock	Life	Vibration
Humidity	Temperature rise	Fungus
Explosion	Terminal torque	Salt spray

4.5 Quality conformance inspections.- Quality conformance inspections shall consist of the individual inspections.

4.5.1 Individual inspections.- Each transformer shall be subject to:

(a) Examination of product (4.6.1)
(b) Operation (4.7.1)

4.6 Examinations.-

4.6.1 Examination of product.- Each transformer shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and marking.

4.7 Test methods.-

4.7.1 Operation.- Each transformer shall be tested at rated input voltage to determine compliance with the requirements specified herein and on the applicable MS or drawing, employing the intended lamp load or its equivalent resistive load. The input voltage shall be as specified at a frequency of 400 cps, ±5 percent. (See 3.6.1.)

4.7.2 Voltage regulation.- The voltage shall be measured using a properly calibrated voltmeter after the transformer has reached a constant temperature with the rated input voltage and lamp or resistive load. The test shall be repeated with the input voltage 10 percent below and above the rated voltage. (See 3.6.2.)

4.7.3 Stop strength.- A tensile load of 25 pounds shall be applied to the stops normal to the axis of the control shaft for at least 1 minute. (See 3.6.3.)

4.7.4 Shaft torque.- The torque required to rotate the control shaft (see 3.6.4) shall be determined by any method satisfactory to the procuring activity.

4.7.5 Terminal torque.- A tensile load of 5 pounds shall be applied to the terminals in line with the control shaft axis for at least 1 minute, and then shall be applied in a plane normal to the shaft axis for at least 1 minute. (See 3.6.5.)

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4.7.6 Dielectric strength.- A potential of 1,000V rms, 60 cps shall be applied between the transformer housing and all terminals for 1 minute. (See 3.6.6.)

4.7.7 Overload.- Either one of the following tests shall be met without permanent distortion or damage to the transformer (see 3.6.7):

(a) An overload of 250 percent above the rated maximum current shall be applied cyclicly for 1 minute "on" and 19 minutes "off" for 100 hours.

(b) An overload of 150 percent above the rated maximum current shall be applied cyclicly for 5 minutes "on" and 15 minutes "off" for 100 hours.

4.7.8 Life.- The transformer shall be subjected to its maximum rated resistive load for 500 hours at an ambient temperature of 52° C. Readings shall be taken every 24 hours to determine compliance with 3.6.11.(See 3.6.8.)

4.7.9 Cycling.- The transformer shaft shall be rotated from the "off" position to the maximum clockwise position and returned to the "off" position for 10,000 cycles. During this test, the rated input voltage and resistive load shall be applied. (See 3.6.9.)

4.7.10 Temperature rise.- Readings for temperature rise shall be taken with the transformer delivering its maximum rated load for at least 30 minutes in an ambient temperature of 25° C. (See 3.6.10.)

4.7.11 Volt-ampere output.- The transformer shall be tested to determine that it meets the requirements of figure 1 and 3.6.11.

4.7.12 Environmental.- The transformers shall be subjected to the following tests conducted in accordance with the specified procedures of MIL-E-5272.

4.7.12.1 High temperature.- The transformer shall be subjected to high temperature in accordance with Procedure II. (See 3.6.12.)

4.7.12.2 Low temperature.- The transformer shall be subjected to low temperature in accordance with Procedure I. (See 3.6.12.)

4.7.12.3 Humidity.- The transformer shall be subjected to humidity in accordance with Procedure I. (See 3.6.13.)

4.7.12.4 Altitude.- The transformer shall be subjected to altitude in accordance with Procedure VI, condition E. At the end of the 1-hour test period, with the altitude and temperature conditions maintained, the volt-ampere output shall be met. (See 3.6.14.)

4.7.12.5 Vibration.- The transformer shall be subjected to vibration in accordance with Procedure XIII. During the test, the output voltage shall be monitored by an oscilloscope for voltage interruption. Any interruption during the test shall constitute a failure. Upon completion of this test, the transformer shall be tested to determine compliance with 3.6.6. (See 3.6.15.)

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4.7.12.6 Acceleration.- The transformer shall be subjected to acceleration in accordance with Procedure I. (See 3.6.16.)

4.7.12.7 Fungus.- The transformer shall be subjected to fungus in accordance with Procedure I. After completion of this test, the transformer shall show no appreciable growth of fungi. (See 3.6.17.)

4.7.12.8 Sand and dust.- The transformer shall be subjected to sand and dust in accordance with Procedure I. (See 3.6.18.)

4.7.12.9 Temperature shock.- The transformer shall be subjected to temperature shock in accordance with Procedure I. (See 3.6.19.)

4.7.12.10 Explosion.- The transformer shall be subjected to an explosion test in accordance with Procedure III. (See 3.6.20.)

4.7.12.11 Salt spray.- The transformer shall be subjected to salt spray in accordance with Procedure I. (See 3.6.21.)

4.7.13 Radio interference.- The transformer shall be tested for radio interference in accordance with MIL-I-6181. (See 3.6.22.)

4.7.14 Inspection for delivery.- The transformer shall be inspected to determine that the preservation, packaging,packing, and marking are in accordance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking.- Transformers shall be preserved and packaged in accordance with level A or C of MIL-E-17555, as specified (see 6.2). They shall be packed in accordance with level A, B, or C, as specified (see 6.2), and marked in accordance with MIL-E-17555. When specified (see 6.2), a rough handling test in accordance with MIL-P-7936 shall be conducted in lieu of the test required in MIL-E-17555.

6. NOTES

6.1 Intended use.- The transformer is intended for use in controlling aircraft lighting circuits which operate from a nominal 400-cycle source at the specified input and output voltage.

6.2 Ordering data.- Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) MS part number (see 3.5).
(c) Where the preproduction inspections are to be conducted (see 4.4.1).
(d) Applicable level of preservation, packaging, and packing (see 5.1).
(e) Data requirements (see 3.2).
(f) When the drop test shall conform to MIL-P-7936 (see 5.1).

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6.3 Definitions.- For the purpose of this specification, the following definitions will apply.

6.3.1 Transformer.- Transformer is defined as an autotransformer or a step-down transformer with a variable output control.

6.3.2 Voltage regulation.- Voltage regulation is defined as the no-load voltage minus the full-load voltage over no-load times 100.

6.3.3 Cycle of operation.- A cycle of operation is defined as the rotation of the shaft from the "off" position to the maximum clockwise position and return with rated input voltage and connected resistive load.

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Air Force - (11)

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